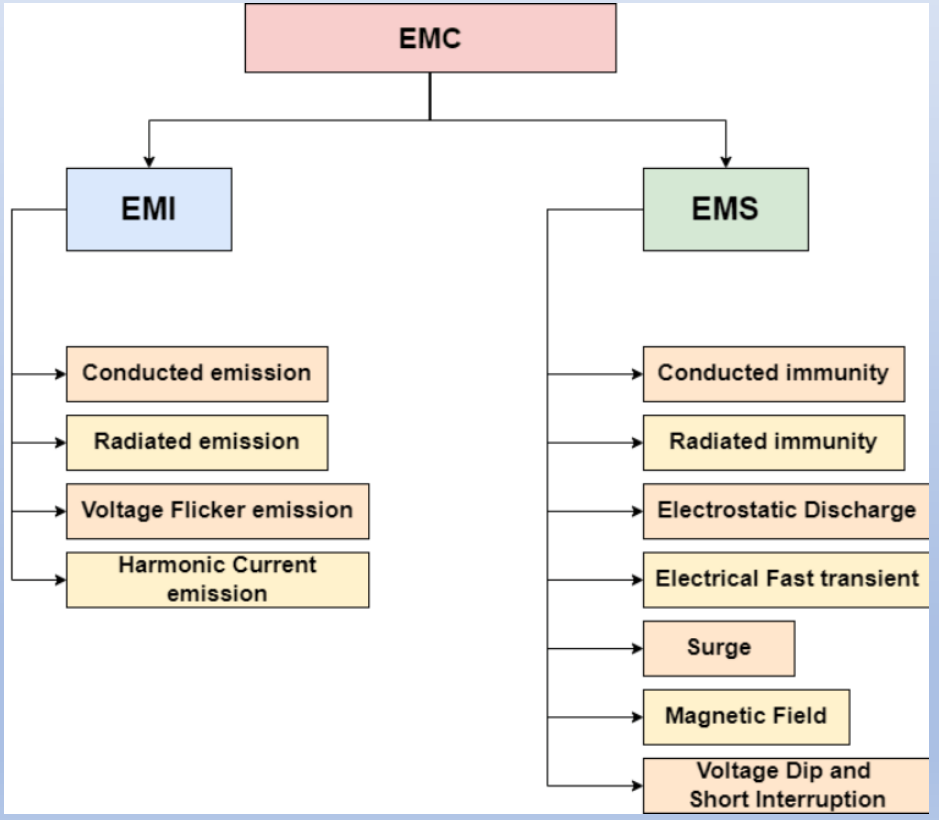
Machine learning measures for EMC solutions

A Multi-Model Approach to Electromagnetic Compatibility

Markosyan Arman

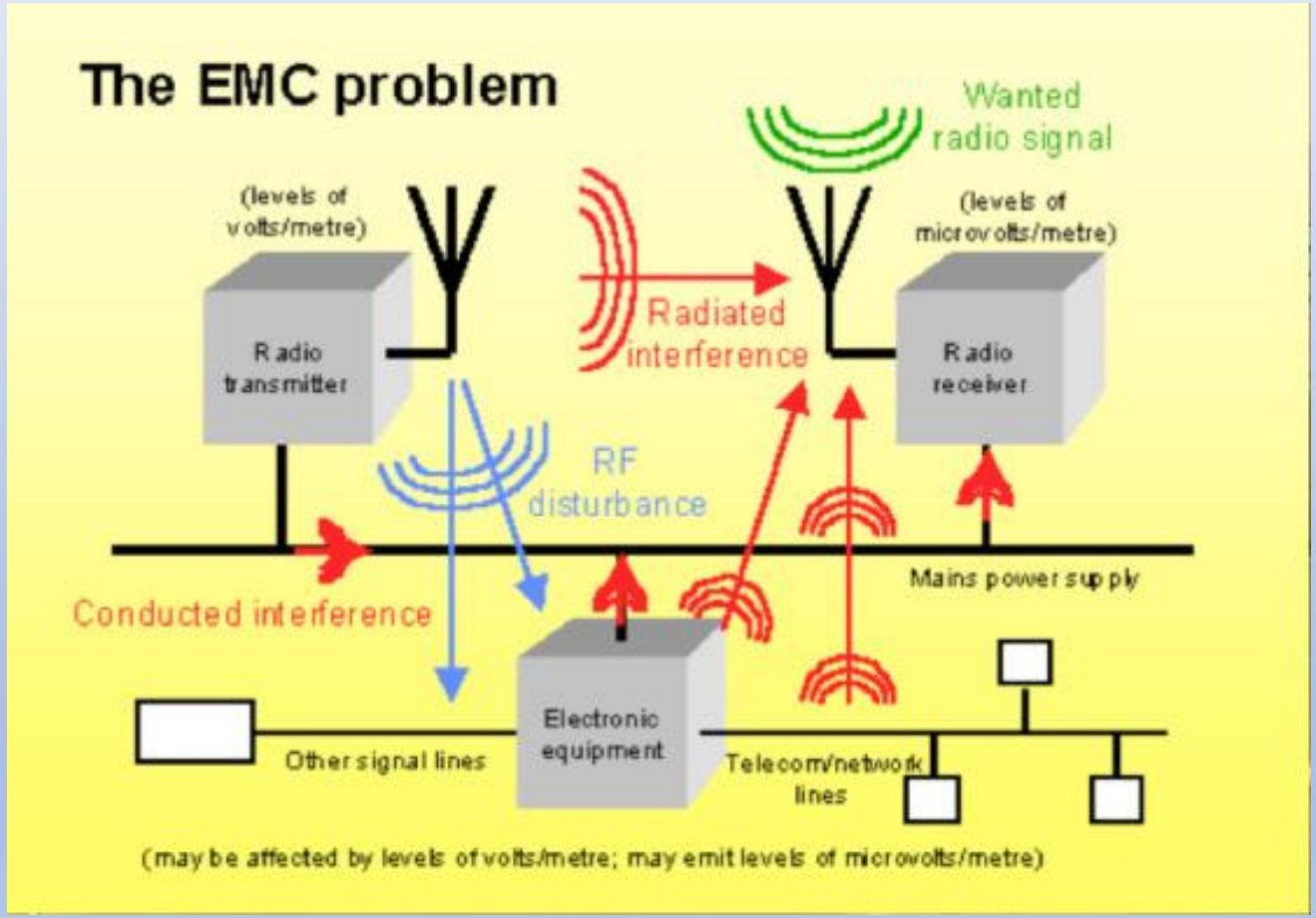


The Challenge in EMC

The Problem: Traditional solvers (FEM, SPICE) are slow and computationally expensive. Pure Deep Learning ignores physics laws.

Solution:

Space: Predicting IR-Drop on chips instantly.
Time: Modeling Crosstalk in transmission lines.
Physics: Simulating Wave Propagation without data.



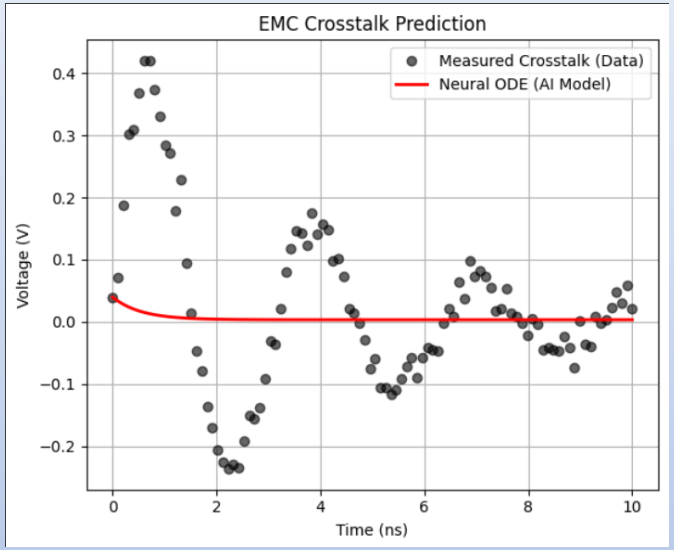
Model 1 Spatial Operator Learning (FNO)

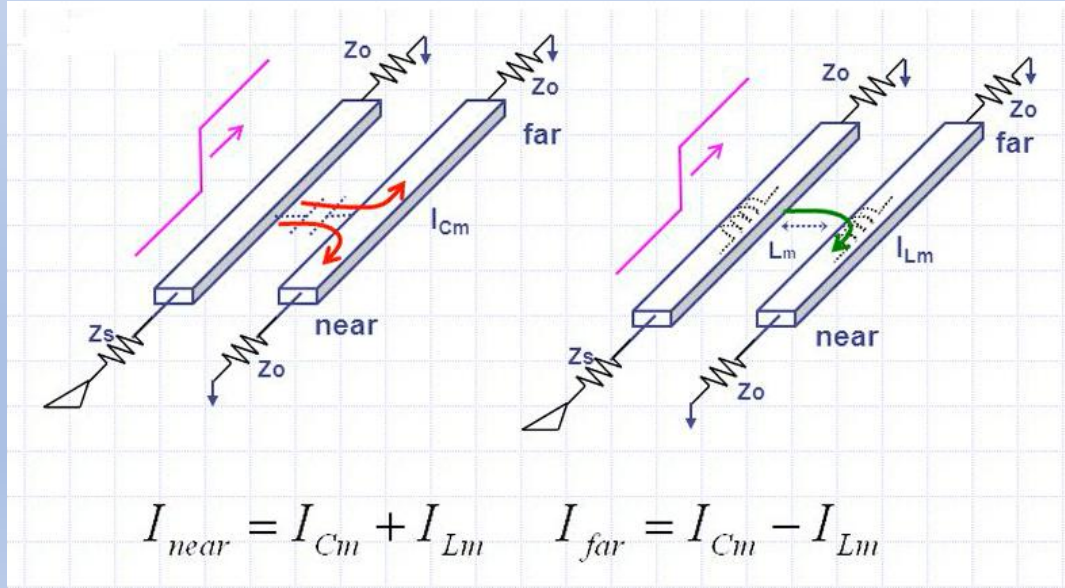
- Problem: Conducted Emissions (IR-Drop) in VLSI Circuits.
- Method: Fourier Neural Operator (FNO).
  Learns the mapping from Geometry (Rudy density) to Voltage Field globally in the frequency domain.

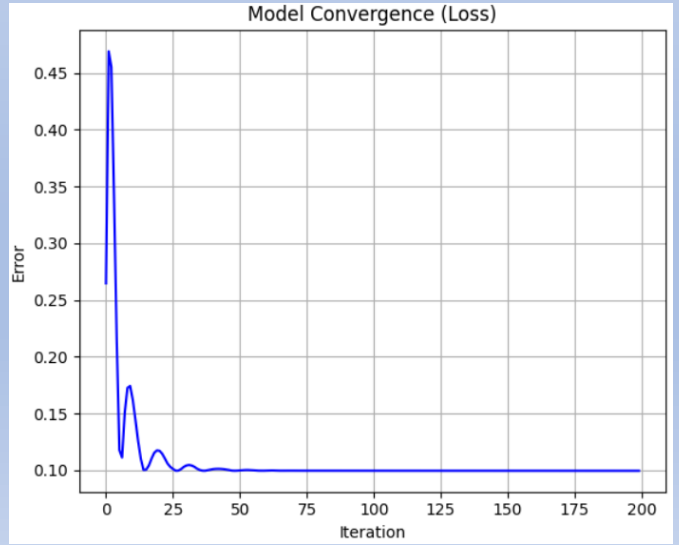
Model 1 Spatial Operator Learning

Model 2 Temporal Dynamics (Neural ODE)

- **Problem:** Crosstalk & Signal Integrity in Transmission Lines.
- **Method:** Neural Ordinary Differential Equations (Neural ODE).
- Goal: To create a model capable of learning the underlying differential equation from noisy measurements.

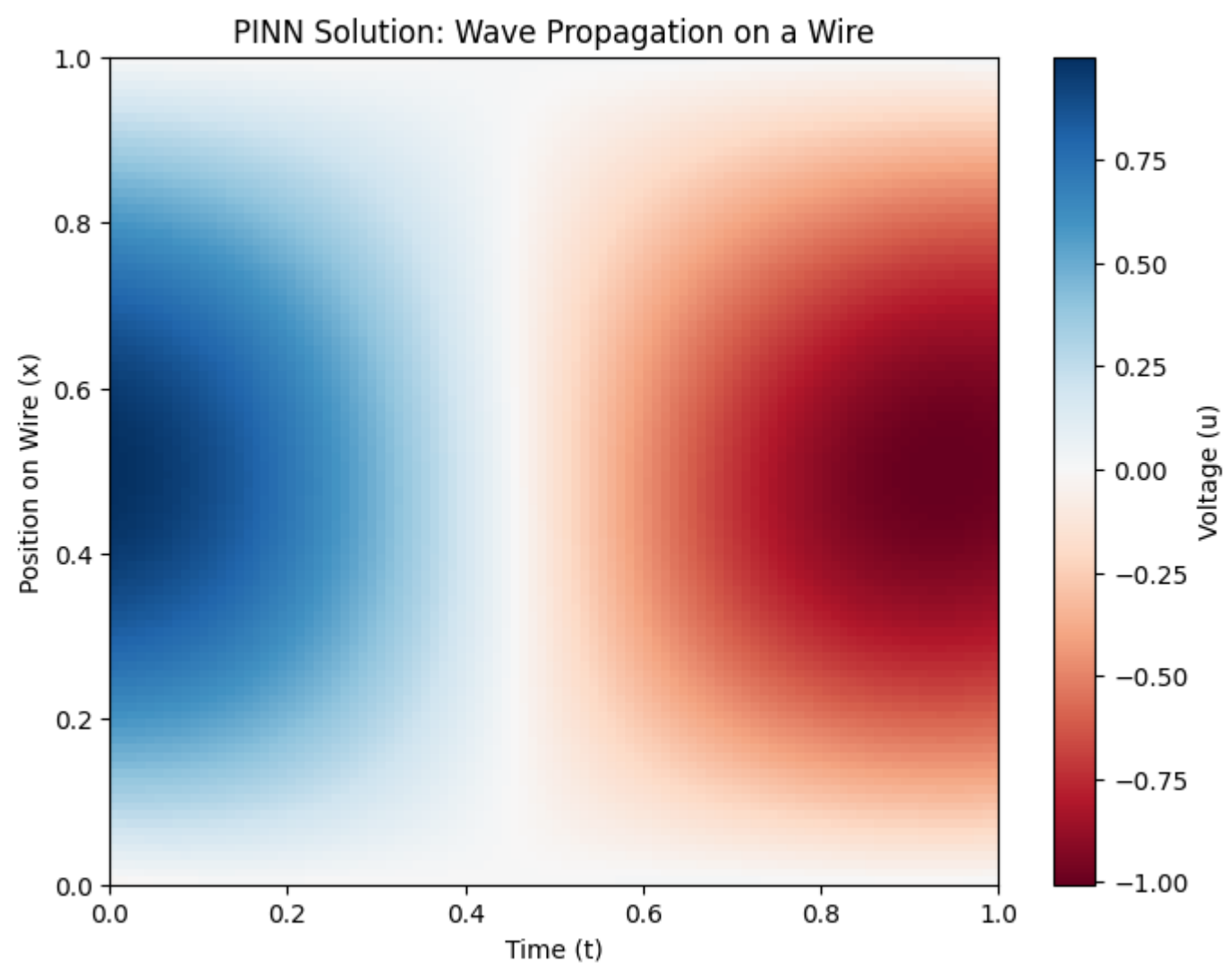






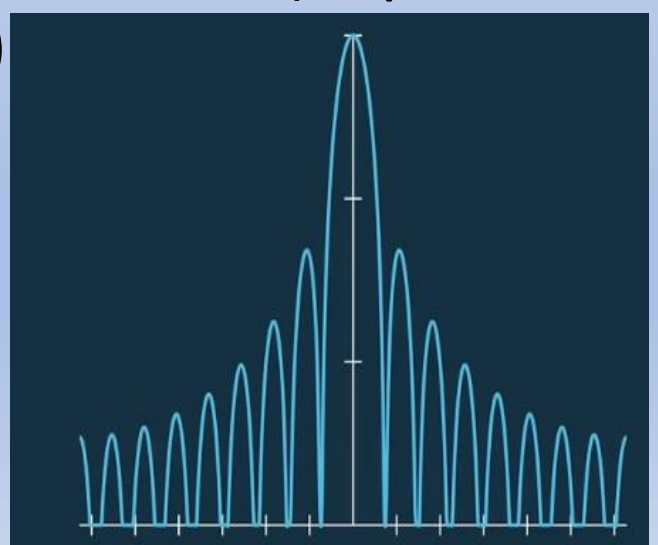
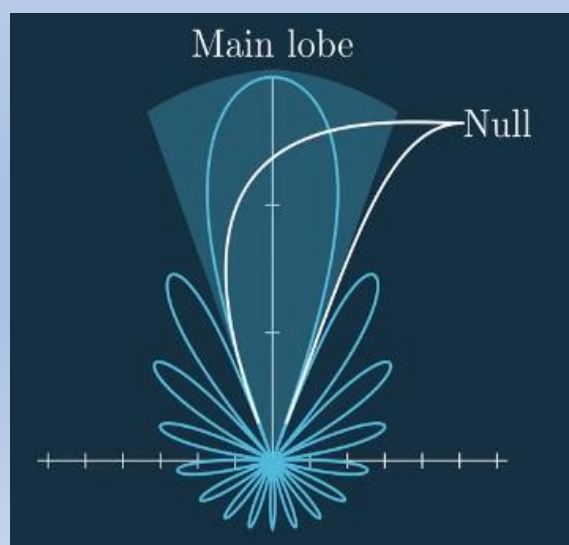
Model 3 using the PINN method

- Fundamental Wave Propagation (1D Wave Equation)
- PINN Minimizes residual error



Comparative Analysis of Models

- Model 1
  - Focus: Chip Geometries & Voltage Maps (CircuitNet)
  - Learns in the frequency domain
  - Predicts full-field distributions
- Model 2
  - Focus: Signal Integrity & Crosstalk (NASA Data)
  - Learns the continuous derivative
  - Handles noisy, irregular sensor data
- Model 3
  - Focus: Fundamental Wave Propagation.
  - Minimizes residual error of the Wave Equation

Practical Impact & Future Work (HF)

- Accelerates EMC compliance testing
- Reduces reliance on expensive commercial solvers
- Combine PINN with FNO (Physics-Informed Operators).
- 3D Enclosure Shielding
- Antenna radiation calculation and simulation (implementation of models in the high-frequency range)
- Practical Implementation